

Exercise 1: Motion Recording Analysis

The figure below represents the recording of the different positions A_i taken by a self-propelling mobile set at $\tau = 40$ ms.

Given: $A_0A_1 = 1.2$ cm; $A_1A_2 = 2$ cm; $A_2A_3 = 2.8$ cm; $A_3A_4 = 3.6$ cm; $A_4A_5 = 4.4$ cm.



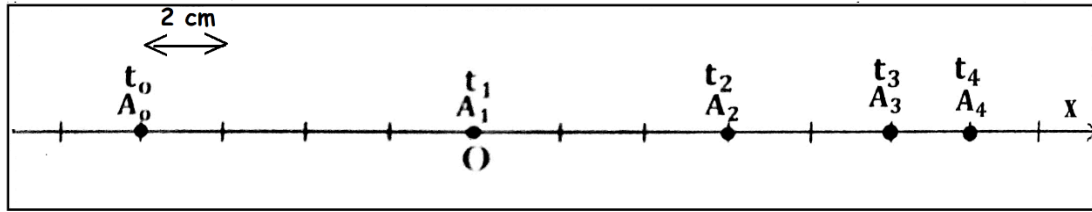
1. The movement of the mobile is rectilinear. Justify.
2. The movement of the mobile is accelerated. Justify.
3. What is the abscissa of the mobile at $t_0 = 0$ and at $t_1 = 40$ ms?
4. Consider the table below:

A_i	A_0	A_1	A_2	A_3	A_4	A_5	A_6
t_i (ms)	0	40	80	120	160	200	240
V_i (m/s)		0,4	v_2	v_3	1	1,2	

- 4.1. Calculate the instantaneous speeds v_2 and v_3 .
- 4.2. Determine in the coordinate system $(O, \vec{i})$, the characteristics of the velocity vector $\vec{v}_1$.
- 4.3. Represent using the scale: $1 \text{ cm} \leftrightarrow 0.2 \text{ m.s}^{-1}$, the vector $\vec{v}_1$.
5. Calculate the instantaneous accelerations a_2 and a_4 .
6. Determine in the coordinate system $(O, \vec{i})$, The characteristics of the acceleration vector $\vec{a}_2$.
7. Represent using the scale: $1 \text{ cm} \leftrightarrow 0.2 \text{ m.s}^{-2}$, the vector $\vec{a}_2$.
8. Knowing that the acceleration keeps constant over time, calculate V_6 .

Exercise 2: Motion Recording Analysis

Particle (A) is in rectilinear motion on the x-axis. The figure below shows successive positions of (A) separated by a time interval $\tau = 40 \text{ ms}$.

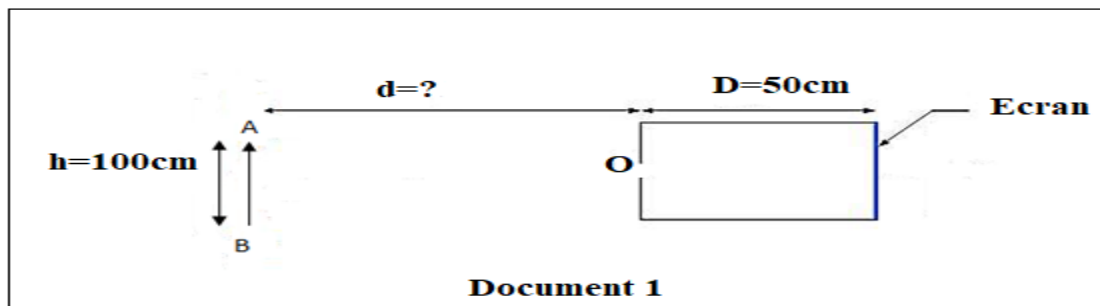


1. Indicate the positions occupied by particle (A) at t_0 , t_1 , t_2 , t_3 et t_4 .
2. What is the displacement of (A) during $\Delta t = t_1 - t_0$.
3. Calculate the average speed $V_{0,4}$ of the particle (A) between the dates $t_0 = 0$ and $t_4 = 4\tau$.
4. Calculate the instantaneous speeds V_1 , V_2 and V_3 .
5. Determine in the coordinate system $(O, \vec{i})$, the characteristics of the velocity vector $\vec{v}_2$.
6. Represent using the scale $1 \text{ cm} \leftrightarrow 1.25 \text{ m/s}$, the velocity vector. $\vec{v}_2$.
7. Calculate the algebraic value of the acceleration vector $\vec{a}_2$.
8. Write the expression of $\vec{a}_2$.
9. Represent using the scale $1 \text{ cm} \leftrightarrow 10 \text{ m/s}^2$, the acceleration vector $\vec{a}_2$.
10. What's the nature of motion of (A) based on the representation of vectors $\vec{v}_2$ and $\vec{a}_2$.

Exercise 3: Rectilinear propagation of light

A black chamber is a simple setup that allows the formation of the image of an object on the screen. Light rays enter this chamber through a small hole O.

In front of this black chamber, at a distance d from O, is placed an object of height $h = 100 \text{ cm}$. The image's height is 10 cm .

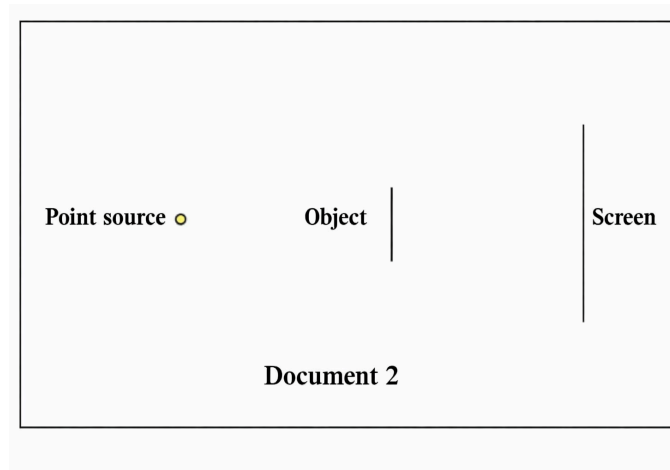


1. Redraw the figure of document 1, then trace the light rays coming from A and B that allow you to draw the image $A'B'$ given by this black chamber. .
2. Determine the distance d separating the object from the black chamber. .

Exercise 4: Shadow of an object

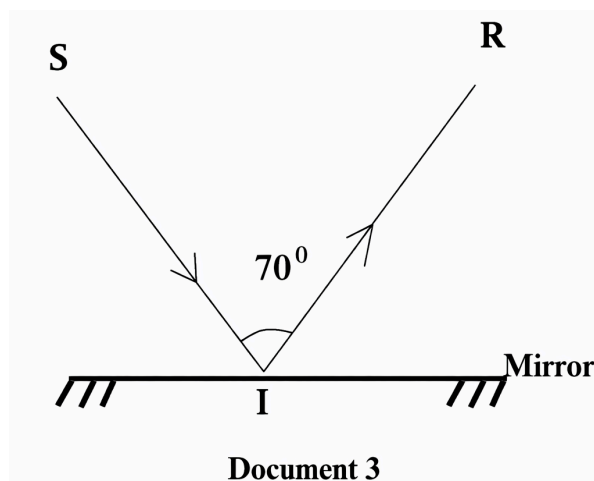
Document 2 shows a point source of light, an object, and a screen.

1. Draw the figure of the document to then trace the light rays, allowing you to obtain the shadow of this object on the screen.
2. Calculate the size of the shadow, knowing that:
 - The size of the object is 15 cm.
 - The distance between the source and the object is 1 m.
 - The distance between the source and the screen is 3 m.
3. To increase the size of the shadow, should we get the source closer or further from the screen?



Exercise 5: Reflection of light

A light ray (SI) falls on a plane mirror. It reflects along (IR) as shown in Document 3 .



The angle between the incident and the reflected ray is 70° .

1. Redraw the figure and trace the normal at the point of incidence I.
2. Calculate the angle of incidence $\hat{i}$. Deduce the reflection angle $\hat{r}$.